ALEXANDER JAMES WRIGHT

Cambridge, United Kingdom

alex.wright@gmail.com

+44 7911 482756

LinkedIn: linkedin.com/in/alexjwright

GitHub: github.com/alexjwright

PROFESSIONAL SUMMARY

Machine Learning Engineer with 5 years of experience building and deploying production ML systems. I've worked across computer vision, NLP, and recommendation systems in both startup and scale-up environments. I'm particularly interested in making models faster and more interpretable rather than just chasing benchmark scores. Outside of work, I contribute to open-source ML projects and enjoy writing technical blog posts about engineering challenges I've encountered.

TECHNICAL SKILLS

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM

Computer Vision: OpenCV, YOLO, ResNet, image classification, object detection

Natural Language Processing: NLTK, spaCy, transformer models, BERT

Data Engineering: SQL, Pandas, Apache Spark, data pipelines

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform

Tools and Frameworks: Docker, Kubernetes, Git, Jupyter, MLflow

Languages: Python, Java, SQL, Bash

WORK EXPERIENCE

Senior Machine Learning Engineer

Quantum Insights Ltd

www.quantuminsights.io

Cambridge, United Kingdom

March 2022 - Present

Designed and implemented a real-time fraud detection system that reduced false positives by 34% and now processes 2 million transactions daily using gradient boosting ensemble methods. Led the migration of legacy scikit-learn models to a TensorFlow-based architecture deployed on Kubernetes, cutting inference latency from 850ms to 120ms. Mentored 2 junior engineers on production ML best practices, code review processes, and debugging distributed training pipelines. Collaborated with the data engineering team to build automated feature engineering pipelines that reduced model development time by 40%.

Machine Learning Engineer

DataFlow Systems

www.dataflowsystems.co

Cambridge, United Kingdom

July 2020 - February 2022

Developed a recommendation engine for e-commerce that increased click-through rates by 23% using collaborative filtering and content-based approaches with PyTorch. Built and maintained 5 NLP models for customer intent classification achieving 94% F1 score in production with latency under 200ms. Implemented comprehensive monitoring and alerting for model drift using statistical tests, catching 3 critical performance

degradations before they impacted users. Contributed to MLOps infrastructure by creating reusable training pipelines and model serving templates using Docker and Jenkins CI/CD.

Junior Machine Learning Engineer

Vertex Analytics

www.vertexanalytics.net

Cambridge, United Kingdom

June 2019 - June 2020

Contributed to computer vision projects involving object detection and image segmentation for industrial quality assurance systems with 92% average accuracy. Processed and cleaned large datasets containing over 100,000 images, implementing data augmentation strategies to improve model generalization. Participated in sprint planning and agile ceremonies while learning production code standards and version control workflows. Assisted in documenting machine learning methodology and results for client-facing reports and internal knowledge base.

EDUCATION

Master of Science in Machine Learning and Adaptive Intelligence

University of Edinburgh

Edinburgh, United Kingdom

Graduated: August 2019

Distinction

Bachelor of Science in Computer Science with Mathematics

University of Cambridge

Cambridge, United Kingdom

Graduated: June 2018

Upper Second Class Honours

PUBLIC REPOSITORIES

github.com/alexjwright/time-series-anomaly-detector - Unsupervised anomaly detection library for time series data using isolation forests and autoencoders, 340 stars

github.com/alexjwright/interpretable-nlp-models - Collection of tools for visualizing attention mechanisms and extracting feature importance from BERT models, 210 stars

github.com/alexjwright/ml-monitoring-dashboard - Real-time dashboard for tracking model performance metrics and data drift across multiple production models, 185 stars

github.com/alexjwright/efficient-image-preprocessing - GPU-accelerated image preprocessing pipeline for large-scale computer vision projects using CUDA and OpenCV, 128 stars

CERTIFICATIONS

AWS Certified Machine Learning Specialty

Issued: November 2021

Professional Data Engineer - Google Cloud Platform

Issued: May 2021